Question Bank Unit Wise
Data Mining & Analytics (23CB605T(iii))

Unit 1

1. Explain the concept of Data Mining and its importance in modern data analysis.
2. Discuss OLAP & KDD.
3. Explain the relationship between Data Mining and related technologies such as Machine Learning, DBMS, and Statistics.
4. Explain the stages involved in the Data Mining process with suitable explanation.
5. Explain different Data Mining techniques used for extracting knowledge from large datasets.
6. Explain various applications of Data Mining in real-world domains.
7. A retail company wants to analyze its large customer transaction database to discover buying patterns using data mining techniques. Apply the architecture of a data mining system and explain how each component (database, data warehouse server, data mining engine, pattern evaluation module, and user interface) will work together in this scenario.
8. Discuss various uses of Data Preprocessing.

Unit 2

1. Explain the concept of Data Preprocessing and its importance in Data Mining.
2. Explain the different techniques used in Data Cleaning.
3. Explain Data Transformation and Data Reduction with suitable examples.
4. Explain the process of Discretization and generating concept hierarchies.
5. Explain different Data Visualization techniques used in Data Mining.
6. Apply appropriate data preprocessing techniques to improve the quality of a dataset containing missing and inconsistent values.
7. Analyze the role of data cleaning in improving the accuracy of data mining results.
8. Apply data transformation methods to convert raw data into a suitable format for data mining.
9. Analyze how data reduction techniques help in handling large datasets efficiently.
10. Apply discretization techniques to convert continuous data into categorical data with a suitable example.
11. Analyze the importance of concept hierarchies in knowledge representation.
12. Apply data visualization techniques to represent patterns and trends in a dataset.
13. Analyze the differences between data cleaning, data transformation, and data reduction in data preprocessing.
14. Apply appropriate data preprocessing techniques to improve the quality of a dataset containing missing and inconsistent values.

Unit 3

1. Analyze the differences between association rule mining and classification algorithms in data mining
2. Analyze the working of decision tree algorithms for classification and explain how they help in knowledge discovery. ***(BT Level 4)***
3. Apply classification techniques to categorize data into different classes using decision trees. Explain the procedure. ***(BT Level 3)***
4. **Minimum Support = 2**

Find Frequent Itemsets using Apriori Algorithm.

TID	Items Purchased
T1	Bread, Milk
T2	Bread, Diaper, Beer, Eggs
T3	Milk, Diaper, Beer, Coke
T4	Bread, Milk, Diaper, Beer
T5	Bread, Milk, Diaper, Coke

5. **Find Frequent Itemset using Apriori Algorithm**
   Minimum Support = 2

TID	Items
T1	A, B, C
T2	A, C
T3	A, D
T4	B, C
T5	A, B, C

6. **Given Transaction Database**

TID	Items
T1	A, B
T2	B, C, D
T3	A, C, D
T4	A, B, C
T5	A, B, C, D

Minimum Support = 2
Find frequent itemsets using FP-Growth Algorithm.

7. **Numerical on Decision Tree (5 Marks)**
   Given Dataset

Outlook	Play Tennis
Sunny	No
Sunny	No
Overcast	Yes
Rain	Yes
Rain	Yes
Rain	No
Overcast	Yes
Sunny	No

Construct a Decision Tree for classification.

8. Given Transaction Database

TID	Items
T1	A, B, C
T2	A, C
T3	B, C
T4	A, B
T5	A, B, C

Minimum Support = 2

Find frequent item sets using ECLAT algorithm.